

A NEONATAL EMERGENCY RESPONSE SWADDLE (NERS) FOR NEWBORN RESUSCITATION IN LOW- RESOURCE SETTINGS

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ABSTRACT

Birth asphyxia is the leading cause of neonatal mortality in Kenya, accounting for 31.6% of all newborn deaths. Kenya also recorded approximately 31,000 neonatal deaths in 2024 at a rate of 20.7 per 1,000 live births. The critical window for intervention, termed the Golden Minute, is the sixty seconds following birth during which every 30-second delay in initiating ventilation increases mortality risk by 16%. In low-resource public health facilities such as Thika Level 5 Hospital, current resuscitation approaches require birth attendants to coordinate three separate interventions; thermal regulation, airway clearance, and chest stimulation, using three separate pieces of equipment, under extreme time pressure and often with limited skilled personnel. This fragmented approach introduces procedural delays that cost lives.

This report presents the design of the Neonatal Emergency Response Swaddle (NERS), a single integrated device that simultaneously delivers all three primary resuscitation functions to a non-breathing newborn immediately at birth. The NERS is a swaddle-form device incorporating: embedded heating strip and an MLX90614ESF infrared temperature sensor for automated thermal regulation; a 28BYJ-48 stepper motor in a peristaltic pump configuration for controlled airway suction; and two additional 28BYJ-48 stepper motors with padded compression elements for rhythmic chest stimulation. All three functions are individually activatable via labelled push buttons and their status is communicated in real time through a 0.96-inch SSD1306 OLED display. The system is controlled by an Arduino Nano V3.0 microcontroller and key structural components are 3D-printed for low-cost, lightweight construction.

The device was designed specifically for the low-resource healthcare context in Kenya, with a total bill of materials of KES 5,937, making it economically viable for deployment at Level 4 and 5 public facilities. Prototype testing on a neonatal mannequin confirmed correct operation of all three modes, temperature monitoring with real-time OLED feedback, and stable stepper motor performance. Future work includes integration of continuous neonatal heart rate monitoring, chest pad biomechanical optimization, and clinical validation in a simulated delivery room setting.

Keywords: *Neonatal resuscitation, birth asphyxia, neonatal emergency, thermal regulation, airway clearance, chest compression, Arduino, low-resource settings, Kenya, biomedical device*

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ABBREVIATIONS AND ACRONYMS

Abbreviation	Full Form
AHA	American Heart Association
APGAR	Appearance, Pulse, Grimace, Activity, Respiration score
BMV	Bag-and-Mask Ventilation
DHS	Demographic and Health Survey
GPIO	General Purpose Input/Output
ILCOR	International Liaison Committee on Resuscitation
I2C	Inter-Integrated Circuit (communication protocol)
KDHS	Kenya Demographic and Health Survey
KHIS	Kenya Health Information System
LED	Light Emitting Diode
LMIC	Low- and Middle-Income Country
NERS	Neonatal Emergency Response Swaddle
NMR	Neonatal Mortality Rate
NRP	Neonatal Resuscitation Program
OLED	Organic Light-Emitting Diode
PPV	Positive Pressure Ventilation
PVC	Polyvinyl Chloride
SDG	Sustainable Development Goal
SSD	Solomon Systech Display (OLED driver)
UNICEF	United Nations Children's Fund
WHO	World Health Organization

CHAPTER ONE: INTRODUCTION

1.1 Research Background

The first sixty seconds after birth, referred to in neonatal medicine as the Golden Minute, represent the most critical window in a newborn's extrauterine transition. The vast majority of newborns make this transition without assistance. However, a significant proportion do not, and when they fail to breathe, the speed and quality of clinical response in that first minute is the primary determinant of whether they live, die, or survive with permanent neurological damage.

Birth asphyxia, defined by the World Health Organization as the failure to initiate and sustain breathing at birth, is the single leading cause of neonatal mortality in Kenya, accounting for 31.6% of all newborn deaths according to UNICEF [1]. Kenya recorded approximately 31,000 neonatal deaths in 2024, at a rate of 20.7 per 1,000 live births, a rate that has declined by only 22% since 1990 [2]. This slow progress in neonatal survival is a documented characteristic of the sub-Saharan African region, where the neonatal mortality rate of 28 per 1,000 live births is significantly above the global average [3].

The Kenya Demographic and Health Survey (KDHS) 2022 confirmed that neonatal mortality rate stood at 21 deaths per 1,000 live births, with birth asphyxia, neonatal sepsis, respiratory distress, and hypothermia identified as the leading causes among preterm and low birth weight infants [4]. Aluvaala et al. (2021), in a multisite retrospective cohort study of 21 Kenyan hospitals, documented that neonates account for 46% of pediatric admissions but 66% of pediatric deaths, a stark disproportion that reflects the vulnerability of the newborns [5].

The time-criticality of neonatal resuscitation is quantified in the literature. Evidence from Mwangi et al. (2018), derived from direct observation of resuscitations at Kakamega County General Hospital, Kenya, demonstrates that the risk of death increases by 16% for every 30-second delay in initiating ventilation during the first six minutes following birth [6]. This figure, a 16% increase in mortality risk for every half-minute of delay, frames the entire engineering challenge that motivated this project.

Compounding the time-critical nature of the problem, is the multi-task burden placed on the birth attendant in a low-resource setting. In high-income settings, neonatal resuscitation involves a trained resuscitation team: one provider manages the airway, one provides chest compressions,

one monitors vital signs, and one coordinates. In most Kenyan public hospitals, these tasks fall to a single nurse or midwife, operating with fragmented equipment under acute pressure.

This reality was directly observed by members of the NERSZ team at Thika Level 5 Hospital, Kiambu County, Kenya, during a clinical immersion session. A neonate was delivered without spontaneous breathing. The single birth attendant present had to sequentially: physically transfer the infant to a resuscitaire; stimulate breathing manually by drying; connect a separate suction catheter to a wall-mounted suction machine to clear mucus from the airway; and then initiate manual chest compressions using two fingers on the sternum. Each transition between tasks consumed time. Time that, by the evidence cited above, directly translates to increased probability of death or long-term neurological injury.

It was this gap, between the theoretical understanding of what neonatal resuscitation requires and the practical reality of how it is delivered in under-resourced settings, that this project set out to address.

1.2 Problem Statement

There is a critical absence of integrated, rapid-deployment, affordable neonatal resuscitation support at the point of delivery in low-resource Kenyan health facilities. Current approaches to managing birth asphyxia in such settings require birth attendants to simultaneously or sequentially administer three separate interventions, which are thermal regulation, airway clearance, and chest stimulation, using three separate pieces of equipment, under conditions of acute clinical pressure, often with limited or single-person staffing.

This fragmentation introduces measurable procedural delays. Each transition between tasks represents lost time within the Golden Minute, and as the literature confirms, every 30 seconds of delay in initiating ventilation increases neonatal mortality risk by 16% [6]. Furthermore, the quality of manual interventions, particularly chest compressions, is inherently operator-dependent. A 2024 PLOS Global Public Health analysis of the 2022 KDHS found that only 32.7% of Kenyan newborns receive all recommended components of quality neonatal care after childbirth, a figure that reflects systemic gaps in the delivery of standardized resuscitation support [7].

Additionally, 17.5% of neonates admitted to 21 study hospitals in Kenya were found to be hypothermic on admission, with hypothermia associated with a 35% increase in odds of inpatient

neonatal death [8]. Heat loss begins immediately after birth when the neonate transitions from the thermoneutral uterine environment to the ambient room temperature.

The problem is therefore twofold: delays caused by the fragmentation of equipment and tasks; and the operator-dependency of intervention quality. Both can, in principle, be resolved through integration and automation. That is precisely what NERS is designed to achieve.

1.3 Research Objectives

1.3.1 General Objective

To design and develop a Neonatal Emergency Response Swaddle (NERS) that integrates thermal regulation, airway clearance, and chest stimulation into a single portable device, deployable by a single birth attendant in under-resourced health facilities at the moment of birth, thereby reducing procedural delays in neonatal resuscitation.

1.3.2 Specific Objectives

- To develop a thermal regulation subsystem using a temperature sensor and embedded heating strips to monitor neonatal body temperature in real time and provide controlled warmth.
- To implement an airway clearance subsystem using a 28BYJ-48 stepper motor in a peristaltic pump configuration with silicon tubing to generate controlled negative pressure for gentle neonatal airway suction.
- To design a chest stimulation subsystem using two 28BYJ-48 stepper motors with padded compression elements to deliver rhythmic, calibrated chest stimulation.
- To validate prototype performance through mannequin testing of all three operational modes, confirming correct mechanical and electronic function prior to any future clinical evaluation.

1.4 Justification

This project is justified on clinical, technical, economic, and national health policy grounds.

From a clinical perspective, the consequences of delayed or inadequate neonatal resuscitation are severe and well-documented. Birth asphyxia survivors who do not receive timely resuscitation face elevated risks of hypoxic-ischemic encephalopathy (HIE), cerebral palsy, epilepsy, cognitive impairment, and developmental delay [9]. Effective resuscitation at birth can prevent

approximately 30% of birth asphyxia deaths and avert 5–10% of deaths due to complications of preterm birth [6]. An integrated, single-device approach that eliminates transition delays directly addresses this mortality burden.

From a technical perspective, the convergence of stepper motor technology, non-contact infrared sensing, and microcontroller-based embedded systems has created a viable pathway for developing a working prototype device within a constrained budget. The 28BYJ-48 stepper motor is widely validated in research prototyping for its precise positional control, low voltage operation (5V), and compatibility with the Arduino ecosystem via the ULN2003 driver board [10]. The MLX90614ESF sensor has demonstrated measurement accuracy within $\pm 0.5^{\circ}\text{C}$ of clinical reference standards in embedded health monitoring applications [11].

From an economic perspective, the device's total cost of KES 5,937 makes it economically feasible for procurement by Level 4 and 5 public health facilities in Kenya. The Ministry of Health Kenya's Newborn and Child Health Strategic Plan 2022–2026 explicitly prioritizes affordable, deployable innovations for improving neonatal health [12].

From a national health policy alignment perspective, Kenya has committed to reducing neonatal mortality to at least 12 per 1,000 live births by 2030 in line with the Sustainable Development Goal (SDG) 3.2 target [13]. With the current NMR at 20.7 per 1,000 live births.

1.5 Scope and Limitations

1.5.1 Scope

This project encompasses the complete design, development, firmware programming, and prototype validation of the NERS device. The scope includes:

- Hardware design and assembly: Integration of three 28BYJ-48 stepper motors with their ULN2003 driver boards, one MLX90614ESF temperature sensor, one SSD1306 OLED display module, three push buttons and all necessary power and signal wiring on a Veroboard circuit.
- Mechanical design: 3D-printed motor casing and peristaltic pump configuration for the airway suction subsystem within a swaddle form factor.
- Firmware development: Arduino C/C++ code for the Arduino Nano V3.0 implementing all three operational modes with independent activation, real-time temperature acquisition

via the MLX90614 library, stepper motor control using the half-step sequence, debounced button input handling, and OLED display management.

- Prototype validation: Functional testing of all three operational modes on a neonatal mannequin, confirming correct mechanical operation and display feedback.

1.5.2 Limitations

This project acknowledges several limitations that define boundaries for future work:

- Clinical validation: The prototype has been tested on a neonatal mannequin only. No clinical trials involving actual neonates have been conducted. Such trials would require ethical approval, biocompatibility certification, and regulatory clearance beyond the scope of this academic project.
- Force calibration: While the stepper motor compression elements provide rhythmic stimulation, precise force calibration against ILCOR-recommended chest compression depths has not been performed. This requires instrumented mechanical testing.
- Suction pressure quantification: The negative pressure generated by the peristaltic pump has not been measured against clinical suction standards. This requires pressure sensor validation.
- Vital signs integration: The current prototype monitors temperature only. It does not include heart rate, oxygen saturation, or respiratory rate monitoring.
- Long-term durability: The prototype has not been subjected to repeated-use endurance testing, sterilization cycle testing, or ingress protection assessment.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of the existing literature relevant to the NERS project, covering the epidemiology of neonatal mortality and birth asphyxia in Kenya and sub-Saharan Africa.

2.2 Neonatal Mortality in Kenya and Sub-Saharan Africa

2.2.1 Regional and Global Context

The global neonatal mortality rate stands at approximately 18 deaths per 1,000 live births, with sub-Saharan Africa carrying a disproportionate burden of 28 deaths per 1,000 live births, which is more than 1.5 times the global average [3]. Globally, prematurity, birth asphyxia, and neonatal sepsis account for close to eight in ten neonatal deaths, with this burden concentrated overwhelmingly in low- and middle-income countries [2]. The International Liaison Committee on Resuscitation (ILCOR) has identified that timely, effective resuscitation immediately at birth is among the highest-impact interventions available to reduce neonatal mortality in resource-limited settings [14].

The Lancet Global Health (2023) published an ILCOR statement noting that most recommendations on cardiopulmonary resuscitation have been developed from the perspective of high-resource settings, with insufficient adaptation to the specific constraints of low-resource environments, which includes limited staffing, fragmented equipment, and absence of resuscitation teams [15]. This represents a documented gap in global neonatal resuscitation policy that directly motivates the design philosophy of NERS.

2.2.2 Kenya-Specific Epidemiology

The 2022 KDHS confirmed a neonatal mortality rate of 21 deaths per 1,000 live births, with no statistically significant change from the 2014 rate of 22 per 1,000 [4]. By 2024, Kenya's absolute number of neonatal deaths had risen to approximately 31,000 annually due to population growth, even as the rate per 1,000 live births declined marginally to 20.7 [2].

Data from the Kenya Health Information System confirms that over 70% of neonatal deaths are attributable to three causes: prematurity, birth asphyxia, and neonatal sepsis [17]. These three

causes are also the primary conditions driving Kenya's disproportionate pediatric hospital mortality: Aluvaala et al. (2021) documented that neonates constitute 46% of pediatric admissions but 66% of pediatric deaths across 21 Kenyan hospitals [5].

The 2022 KDHS also revealed stark geographic and socioeconomic disparities. Only 31% of deliveries in the poorest households had a skilled attendant at birth, compared to 93% in the richest households [1]. This disparity in skilled attendance, directly correlated with the ability to recognize and manage birth asphyxia, shows the urgency of developing devices that aid the birth attendants in resource-constrained settings.

Moreover, a 2024 analysis of the 2022 KDHS by Asiimwe et al. found that only 32.7% of Kenyan newborns received all recommended components of quality neonatal care after childbirth [7]. This figure quantifies the scale of the systems gap that an integrated device like NERS is designed to address at the point of delivery.

2.3 Birth Asphyxia: Pathophysiology and Clinical Significance

2.3.1 Definition and Incidence

Birth asphyxia is defined by the WHO as the failure to initiate and sustain breathing at birth. It is estimated that approximately 10 million babies globally experience breathing difficulties at birth annually [3]. Approximately 700,000 newborns die from birth complications each year, with an estimated 233,000 surviving with moderate-to-severe neurodevelopmental impairment [3]. In Kenya, birth asphyxia accounts for 29–31.6% of neonatal deaths, making it the single leading cause [1, 5].

2.3.2 Pathophysiology

When a neonate fails to breathe at birth, the sequence of pathophysiological events is rapid and predictable. Primary apnea which is characterized by cessation of respiratory effort in response to hypoxic stress, is followed by gasping respirations. If oxygenation is not restored, secondary apnea develops, which is a deeper, more refractory cessation of breathing that does not respond to stimulation alone and requires positive pressure ventilation. The transition from primary to secondary apnea can occur within minutes, and the ability of birth attendants to distinguish between these states in real-time is limited without monitoring equipment [14].

Accompanying respiratory failure, hypoxia drives progressive bradycardia and ultimately cardiac arrest. The neonatal myocardium is relatively tolerant of hypoxia compared to the adult, but sustained oxygen deprivation above approximately 5–8 minutes results in irreversible myocardial and cerebral injury. The neurological consequences include hypoxic-ischemic encephalopathy (HIE), which in its severe form is associated with cerebral palsy, epilepsy, cognitive impairment, and sensory deficits [9].

2.3.3 The Golden Minute and Time-to-Ventilation

A direct observation study by Mwangi et al. (2018) at Kakamega County General Hospital, Kenya, established that the risk of death increases by 16% for every 30-second delay in initiating ventilation up to six minutes, and by 6% for each additional minute of delay in applying bag-and-mask ventilation [6]. This finding provides the quantitative basis for the engineering requirement, which is any integrated resuscitation device must be deployable within seconds of birth, not minutes.

The American Heart Association (AHA) and American Academy of Pediatrics 2023 reaffirmed that the first 60 seconds after birth are the highest-impact intervention window and that delays in initial assessment and intervention are directly correlated with adverse outcomes [18]. The 2025 ILCOR Neonatal Life Support Consensus on Science with Treatment Recommendations further emphasizes that the structure and speed of the initial response are critical determinants of the outcome of the newborn [19].

2.4 Neonatal Hypothermia

2.4.1 Definition, Prevalence, and Mortality Association

The WHO defines neonatal hypothermia as an axillary temperature below 36.5°C in newborns within the first 28 days of life. It is stratified as mild (36.0°C–36.4°C), moderate (32.0°C–35.9°C), and severe (<32.0°C) [20]. Hypothermia is a major contributor to neonatal mortality, acting both as a primary cause of death and as a comorbidity that amplifies the severity of other conditions including birth asphyxia and sepsis.

Wainaina et al. (2024), in a study across 21 neonatal units in Kenya analyzing 58,804 newborns, found that 17.5% arrived hypothermic on admission. The study found that hypothermia at admission was associated with a 35% increase in the odds of neonatal inpatient death. Significant

associations were found between low birth weight, very low APGAR scores, and hypothermia at admission [8]. This study directly informs the thermal regulation design requirements of NERS.

2.4.2 Heat Loss Mechanisms in the Newborn

Newborns are particularly vulnerable to heat loss because of their high surface-area-to-volume ratio, limited subcutaneous fat insulation, immature thermoregulatory mechanisms, and wet skin at birth which dramatically increases evaporative heat loss. The transition from the thermoneutral uterine environment (approximately 37°C) to the ambient delivery room temperature (typically 22–26°C) creates an immediate thermal gradient across which the neonate loses heat [20].

Without active warming from radiant warmers, heated mattresses, skin-to-skin contact, or integrated warming elements, a neonate can develop hypothermia within minutes. Kyokan et al. (2023), reviewing warming devices for neonates in low-resource settings, confirmed that integrated thermal management remains a critical, underprovided component of neonatal resuscitation in facilities where radiant warmers and incubators are unavailable or non-functional [21].

2.4.3 Temperature Monitoring Requirements

Continuous, non-contact temperature monitoring is the clinical standard in neonatal intensive care. The MLX90614ESF infrared thermopile sensor achieves non-contact temperature measurement with factory-calibrated accuracy of $\pm 0.5^{\circ}\text{C}$ and a 17-bit resolution (0.02°C), communicating via I2C protocol. In embedded health monitoring applications using this sensor with an Arduino Nano, temperature differences smaller than $\pm 0.2^{\circ}\text{C}$ from clinical reference thermometers have been reported [11]. The non-contact mode of measurement is clinically advantageous as it eliminates the need for sensor contact with delicate neonatal skin, which is important in a resuscitation context where skin integrity may be compromised.

2.5 Airway Clearance in Neonatal Resuscitation

2.5.1 Clinical Rationale

Upper airway obstruction due to amniotic fluid, mucus, and debris is a major contributor to respiratory failure in neonates who do not breathe spontaneously at birth. In the clinical scenario observed at Thika Level 5 Hospital, airway clearance via suction was a necessary early step in the

resuscitation sequence. Without a clear airway, all subsequent respiratory interventions are ineffective.

Current WHO guidelines and ILCOR recommendations support airway clearance as part of the initial stabilization steps for a neonate who fails to establish spontaneous breathing, though they note that routine suctioning in vigorous newborns is not recommended. For non-vigorous newborns with obvious airway obstruction, the target population for NERS, gentle suction is recommended [18, 19].

2.5.2 Peristaltic Pump Mechanism

Peristaltic pumps generate fluid flow by the sequential compression of a flexible tube by rollers mounted on a rotating head. The compression-relaxation cycle creates a travelling wave of negative pressure that draws fluid through the tube from the inlet to the outlet. This mechanism is preferred in medical applications because the fluid contacts only the inner surface of the tubing, eliminating cross-contamination risk with the pump mechanism.

The 28BYJ-48 stepper motor, has been demonstrated to drive peristaltic pump configurations at flow rates suitable for airway clearing applications [10]. Its precise, controllable rotation enables the birth attendant or the firmware to regulate suction intensity by adjusting motor speed. The 3D-printed pump head that compresses the silicone tubing can be fabricated to the tolerances required for consistent compression depth, ensuring predictable suction generation.

2.6 Chest Compression in Neonatal Resuscitation

2.6.1 Clinical Guidelines

Current ILCOR and AHA/AAP neonatal resuscitation guidelines recommend initiating chest compressions when the heart rate remains below 60 beats per minute despite adequate ventilation [18, 19]. The recommended technique is the two-thumb/encircling-hands method, which consistently achieves greater compression depth, less provider fatigue, and higher accuracy than the two-finger technique based on a 2023 ILCOR scoping review of 29 randomized crossover manikin studies [22].

The recommended compression depth is approximately one-third of the anterior-posterior diameter of the neonatal chest. For a typical term neonate with an AP diameter of approximately

7–8 cm, this corresponds to a compression depth of approximately 2.3–2.6 cm. The recommended compression rate is 90 compressions per minute, coordinated with a 3:1 compression-to-ventilation ratio [19]. A 2022 randomized controlled animal trial by Bruckner et al. comparing compression rates of 90 versus 180 per minute in asphyxiated newborn piglets provided additional evidence supporting the established 90/min rate as physiologically appropriate [23].

2.6.2 Consistency Challenges in Manual Compression

Despite clear guidelines, achieving consistent chest compression quality under emergency conditions is a documented challenge. A single provider managing both airway and compressions, which is the standard of care in most rural Kenyan delivery rooms, cannot maintain consistent depth, rate, and full chest recoil over time. Provider fatigue, the difficulty of maintaining accurate hand positioning while managing other tasks, and the cognitive load of counting under pressure all reduce compression quality.

Automated compression devices have been developed and validated for adult and pediatric resuscitation to address exactly this problem. For neonates, the concept of an automated compression assistance device is less developed but has been explored in research settings, including the BabySaver platform evaluated in Uganda (2025) which demonstrated significantly shorter time to successful resuscitation compared to standard care, along with higher APGAR scores and reduced neonatal morbidity [24]. NERS builds on this conceptual framework at lower cost and in an integrated swaddle form.

2.7 Existing Technologies and Device Landscape

2.7.1 Standard Resuscitation Equipment

Standard equipment at a typical Kenyan Level 5 delivery room for neonatal resuscitation includes: a resuscitaire (radiant warmer with work surface), a bag-valve-mask device for positive pressure ventilation, a wall-mounted suction unit with suction catheter, and a stethoscope. In better-resourced facilities, a pulse oximeter may be available. This equipment requires the birth attendant to be physically mobile between devices, managing each function sequentially or with assistance.

The critical limitation is not the absence of individual technologies, but the absence of integration. No single device currently available for the Kenyan Level 4/5 facility context combines thermal

regulation, airway clearance, and chest stimulation in a form factor deployable by a single provider at the point of delivery without requiring the infant to be transferred between stations.

2.7.2 Commercially Available Integrated Devices

The BabySaver device is a purpose-built bedside resuscitation platform designed for low-resource settings, providing a firm surface for resuscitation, cord clamp capability, bag-valve-mask holder, and basic monitoring. A 2025 pre-post implementation study at Mbale Hospital, Uganda, found that its deployment in combination with the NeoBeat neonatal heart rate monitor significantly improved resuscitation outcomes by increasing bedside resuscitation rates from 9.3% to 45.3% and reducing time to successful resuscitation from 8 minutes to 5 minutes [24]. However, the BabySaver does not integrate automated chest compression or airway suction, and it is expensive.

2.7.3 Research Prototypes

A 2024 review article in ScienceDirect on simulation-based neonatal resuscitation training in low-resource settings noted the critical gap between training quality and device availability. It was identified that even when providers are trained in resuscitation, the absence of integrated equipment at the point of delivery undermines the effectiveness of that training [25]. This aligns directly with the design motivation for NERS, where a trained provider should not have to compensate for equipment fragmentation.

Research into neonatal vital sign monitoring wearables, including the NeoWarm device (2025), which integrates MAX30102 pulse oximetry, MLX90614 temperature sensing, and ECG via an AD8232 frontend on a Bluetooth SoC for neonatal monitoring during kangaroo care, demonstrates the technical viability of such sensors in neonatal healthcare [26]. The NERS project builds on this technical landscape but orients toward active resuscitation rather than monitoring.

2.8 Thermal Sensing Technologies

The MLX90614ESF infrared non-contact thermometer uses a thermopile detector to measure thermal radiation emitted from the surface of an object, providing temperature measurement without physical contact. It operates on the Stefan-Boltzmann principle which states that every object above absolute zero emits thermal radiation proportional to the fourth power of its absolute temperature. The thermopile sensor converts this radiation into a measurable voltage, which is

processed by the sensor's onboard signal chain and outputted as a calibrated temperature value via I2C.

Key specifications relevant to the NERS application include: measurement range of -70°C to $+382^{\circ}\text{C}$ (object) and -40°C to $+125^{\circ}\text{C}$ (ambient); factory calibration accuracy of $\pm 0.5^{\circ}\text{C}$ in the body temperature range; 17-bit ADC resolution providing 0.02°C minimum detectable temperature difference; and I2C communication at up to 400 kHz [27]. The GY-906 module form factor of this sensor provides 5V-compatible operation via an on-board 3.3V regulator, making it directly compatible with the Arduino Nano's 5V supply rail.

Non-contact infrared thermometry has been validated in neonatal clinical contexts as an accurate and practical alternative to contact-based axillary and rectal thermometry, with the additional advantage of not requiring probe insertion or skin contact that could be uncomfortable or harmful in a neonate undergoing resuscitation [21].

2.9 Integrated Neonatal Resuscitation Devices: Research Landscape

The concept of an integrated, portable neonatal resuscitation device has gained increasing attention in biomedical engineering literature, particularly following the renewed focus on reducing neonatal mortality in LMICs under the SDG framework. Schmolzer et al. and the ILCOR NLS Task Force have highlighted that feedback CPR devices for neonatal cardiac arrest are an emerging area requiring further development [19].

The particular innovation of NERS is the swaddle form factor, which enables simultaneous positioning of all three resuscitation modalities on the neonate's body at the moment of birth, eliminating the need for the birth attendant to transfer the infant between devices or to hold sensors and actuators in position manually. This form factor innovation, combined with the price point below USD 50, represents a novel contribution to the design landscape for integrated neonatal resuscitation support in low-resource settings.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter describes the methodology employed in designing, developing, and testing the NERS prototype. It covers the engineering design philosophy, system architecture, component selection rationale, detailed subsystem design for thermal regulation, airway clearance, and chest stimulation, firmware architecture, mechanical and physical design, budget, and project timeline.

3.2 Design Philosophy and Engineering Approach

The NERS device was designed around three core engineering principles derived directly from the clinical problem identified at Thika Level 5 Hospital:

- **Integration:** All three primary resuscitation interventions, including thermal regulation, airway clearance, and chest stimulation which are embedded in a single device. The birth attendant applies one device, not three. This eliminates device-to-device transition time and reduces cognitive load.
- **Simplicity of Operation:** The device must be operable by a single trained birth attendant under acute emergency conditions. The user interface is therefore reduced to three labelled push buttons and one OLED display.
- **Affordability and Deplorability:** The device must be manufacturable within the budget constraints of a low-resource facility and using components available in the Kenyan market. The total bill of materials was constrained to below KES 6,000 , and all components were sourced from a single Kenyan supplier (PixelElectric, Nairobi).

The engineering approach followed a rapid iterative prototyping cycle consistent with the compressed project timeline.

3.3 System Architecture Overview

The NERS system is organized into four functional layers:

- **Sensing Layer:** The MLX90614ESF infrared temperature sensor provides real-time, non-contact measurement of the neonate's surface temperature at a 4-second update interval. This layer provides the feedback for the thermal regulation subsystem.
- **Control Layer:** The Arduino Nano V3.0 serves as the central processing unit. It receives temperature data from the sensing layer, reads button inputs from the user interface layer,

drives the stepper motors in the actuation layer, and manages the display and indicator outputs.

- **Actuation Layer:** Three 28BYJ-48 stepper motors provide the mechanical work of the device. Motor 3 (suction motor) drives the peristaltic pump for airway clearance. Motors 1 and 2 (left and right motors) provide bilateral chest compression stimulation. Heating elements embedded in the swaddle fabric provide thermal regulation.
- **User Interface Layer:** The SSD1306 0.96-inch OLED display provides real-time temperature readout and active mode indication. Three push buttons (BTN1: Warming; BTN2: Suction; BTN3: Massage) enable independent mode activation.

3.4 Component Selection and Rationale

3.4.1 Arduino Nano V3.0 — Microcontroller

The Arduino Nano V3.0 was selected as the central microcontroller for several reasons. Its ATmega328P processor provides 14 digital I/O pins and 8 analog input pins, sufficient to accommodate the 12 stepper motor control pins (4 per motor x 3 motors), 3 button input pins, 1 heating strip pin, and I2C bus shared by the temperature sensor and OLED display. The Arduino's 5V operating voltage is directly compatible with all selected components. The Arduino IDE with its extensive library ecosystem, including the Adafruit MLX90614 and Adafruit SSD1306 libraries, enabled rapid firmware development within the project timeline.

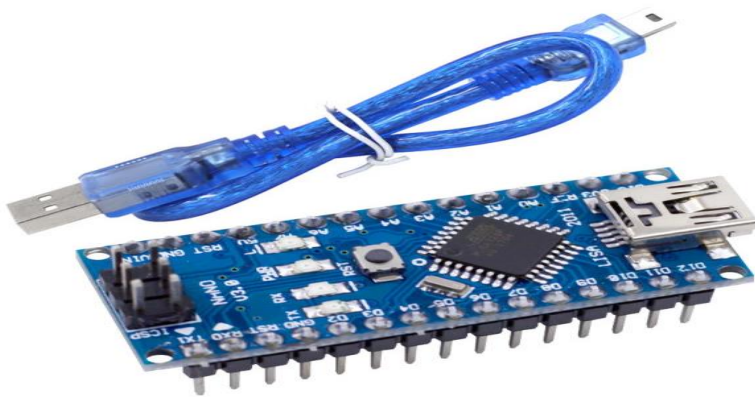


Figure 1 Arduino nano V3.0

3.4.2 28BYJ-48 Stepper Motor — Actuation

The 28BYJ-48 5VDC unipolar stepper motor was selected for all three actuation functions. Its specifications are directly relevant to the NERS requirements. It has 32 internal steps per revolution with a 1:64 gear reduction providing 2048 effective steps per revolution for precise

positional control. Moreover its 5VDC operating voltage matches the Arduino supply, and 240 mA rated current per phase, is within the ULN2003A driver board's capability. Additionally, its dimensions of 28mm diameter x 19mm depth, is compact enough for swaddle integration.

At KES 300 per motor, three motors cost less than USD 7, which is a fraction of the cost of any clinical actuator. The motor's compatibility with the half-step drive sequence implemented in firmware produces smooth rotation necessary for peristaltic pump operation and rhythmic compression delivery.



Figure 2 28BYJ-48 Stepper Motor 5 VDC Unipolar

3.4.3 ULN2003 Stepper Motor Driver Board

The ULN2003A Darlington transistor array driver board provides the current amplification required to drive the 28BYJ-48 motor coils from the Arduino's 5V GPIO pins, which cannot supply sufficient current directly. Each ULN2003 board drives one motor, providing 500 mA per channel, which is more than adequate for the 240 mA motor rating, with flyback diodes integrated for back-EMF protection. At KES 150 per board, three driver boards total KES 450, contributing minimally to the overall budget.



Figure 3 ULN2003 Stepper Motor Driver Board

3.4.4 MLX90614ESF — Temperature Sensing

The MLX90614ESF was selected for its non-contact operation, I2C interface, and accuracy within the clinically relevant neonatal temperature range of 32°C–38°C. Its factory-calibrated accuracy of $\pm 0.5^{\circ}\text{C}$ and resolution of 0.02°C provide the measurement precision required to distinguish normal neonatal temperature of 36.5°C–37.5°C, from mild hypothermia (36.0°C–36.4°C) and from low-grade fever ($>37.5^{\circ}\text{C}$), which are the critical alert thresholds for the NERS warming subsystem. The I2C interface shares the bus with the OLED display, reducing wiring complexity. At KES 1,500 per unit, it is the most expensive single component in the bill of materials, but no lower-cost alternative provides equivalent accuracy and non-contact operation [27].



Figure 4 MLX90614ESF Non-Contact Infrared Thermometer

3.4.5 SSD1306 OLED Display

The 0.96-inch SSD1306 OLED display provides 128x64 pixel resolution with I2C communication at 400 kHz. The OLED technology provides high contrast and wide viewing angles without requiring a backlight, making it readable under the variable lighting conditions of a delivery room. The display is used to show current temperature reading and status (when warming mode is active); active mode indicators ('[WARMING ON]', '[SUCTION ON]', '[MASSAGE ON]'); and button options. The Adafruit SSD1306 library provides a comprehensive graphics API enabling text rendering at multiple sizes and line drawing.



Figure 5 0.96" IIC 128x64 OLED Module

3.4.6 Silicone Pipe — Suction Tubing

Silicone pipe was selected for the peristaltic pump tubing due to its flexibility, chemical inertness, transparency enabling visual confirmation of fluid movement, and compatibility with the compression forces generated by the 28BYJ-48 motor-driven pump head.

3.5 Thermal Regulation Subsystem Design

3.5.1 Design Requirements

The thermal regulation subsystem must:

1. Continuously monitor neonatal surface temperature without skin contact.
2. Provide controlled warmth to prevent hypothermia.
3. Alert the birth attendant when temperature is outside the safe range.
4. Display current temperature in real time on the OLED.

3.5.2 Sensor Configuration

The MLX90614ESF sensor is mounted in the swaddle fabric with its optical window oriented toward the neonate's skin, positioned to measure thoracic surface temperature without contact. The sensor communicates via I2C to the Arduino Nano. The Adafruit_MLX90614 library initializes the sensor and reads object temperature in Celsius.

The firmware reads temperature every 4 seconds, triggered by a time-based condition in the main loop. This sampling interval was selected to provide sufficiently frequent updates to detect temperature trends while minimizing I2C bus activity. The temperature value is stored as a float variable and displayed on the OLED with one decimal place precision.

3.5.3 Temperature Alert Logic

The firmware implements a threshold-based alert system. If the temperature reading falls below 36.0°C, the OLED displays '! LOW' adjacent to the temperature value. If above 37.5°C, it displays '! HIGH'. When within range, 'OK' is displayed. The threshold of 36.0°C corresponds to the WHO definition of the mild/moderate hypothermia boundary, below which active warming intervention escalation is required [20]. The threshold of 37.5°C provides a fever alert threshold appropriate for the immediate newborn period.

3.5.4 Heating Elements

The thermal regulation subsystem includes heating elements embedded within the swaddle fabric. These provide radiant warmth to the neonate's body surface, preventing the evaporative, radiative, and conductive heat losses that occur immediately following birth. The heating elements are activated when Mode 1 (Warming) is enabled via BTN1, toggling the warmingOn flag. In the current prototype, a red LED is activated as a visible indicator when warming mode is on, confirming to the birth attendant that the warming subsystem is active. The OLED displays a 'Baby Warming' confirmation screen for 3 seconds on mode activation before returning to the main home screen.



Figure 6 Warming strip

3.6 Airway Clearance Subsystem Design

3.6.1 Design Requirements

The airway clearance subsystem must:

1. Generate controlled negative pressure at the distal end of a suction catheter.
2. Produce suction that is gentle and controllable, sufficient to remove mucus and fluid from the neonatal airway but not so forceful as to cause airway trauma.
3. Be activatable by a single button press.
4. Operate continuously once activated until the attendant deactivates it.

3.6.2 Peristaltic Pump Configuration

The suction motor which is a 28BYJ-48 stepper motor, controlled via ULN2003 driver board, drives a 3D-printed peristaltic pump head that compresses the silicone tubing. The pump head

consists of a rotor with three rollers positioned to compress the tubing against a curved backing surface. As the rotor turns, each roller compresses and releases a segment of tubing in sequence, generating a travelling compression wave that creates negative pressure behind each roller and positive pressure ahead of it. The inlet side of the tubing is connected to the suction catheter, as the outlet side directs collected fluid to a collection reservoir.

The firmware drives the suction motor using the half-step sequence, advancing one step every SUCTION_STEP_MS milliseconds (1 ms per step). This produces a step rate of 1,000 steps per second, corresponding to approximately 0.49 revolutions per second of the motor shaft (2048 steps/rev at 1ms/step).



Figure 7 Peristaltic suction configuration

3.6.3 Catheter Interface

In the prototype, a neonatal suction catheter can be connected to the inlet end of the silicone tubing. The silicone tube accommodates the compression forces of the pump head while

maintaining adequate flexibility for catheter positioning by the birth attendant. In a clinical implementation, the suction catheter would be sterile and single-use, and the silicone tubing connecting the catheter to the pump would be also autoclavable or also single-use.



Figure 8 Suction Catheter

3.7 Chest Stimulation Subsystem Design

3.7.1 Design Requirements

The chest stimulation subsystem must:

1. Deliver rhythmic, bilateral compression stimulation to the neonatal sternum and lower chest.
2. Operate at a rate consistent with ILCOR neonatal resuscitation guidelines (approximately 90 compressions per minute).

3. Provide controlled, reproducible compression depth.
4. Be activatable by a single button press while the birth attendant maintains other interventions.

3.7.2 Dual Motor Configuration

Two stepper motors via their respective ULN2003 driver boards drive bilateral padded compression elements positioned on either side of the neonatal sternum. The use of bilateral elements distributes compression force symmetrically, reducing the risk of rib stress concentration associated with unilateral point-force compression, while maintaining the two-thumb encircling principle recommended by ILCOR [22].



Figure 9 dual chest stimulation motors

3.7.3 Counter-Rotation Mechanism

The firmware implements a counter-rotation mechanism for the two massage motors to generate a coordinated compressive motion. The left motor steps backward while the right motor steps forward simultaneously, on each cycle of the massage. This counter-rotation causes the two compression elements to move toward each other on the compression stroke and away from each other on the release stroke, producing a squeezing action representative of the two-thumb encircling technique.

Both motors advance one step every `MESSAGE_STEP_MS` milliseconds (1 ms per step), producing the same rotational rate as the suction motor. The compression frequency, the number of full compression-release cycles per minute, is determined by the mechanical geometry of the compression element attachment to the motor shaft, which in the current prototype is set by the physical cam.

3.7.4 Compression Pads

The compression elements consist of padded contacts positioned against the neonatal chest wall within the swaddle fabric. The padding material is selected to provide compliance, distributing contact force over an area rather than a point, while maintaining sufficient firmness to transmit the motor's rotational force as a translational compression. In the prototype, PLA padding is used. In a clinical device, medical-grade silicone padding optimized for neonatal anatomy would be used.



Figure 10 chest stimulation pad against chest wall

3.8 User Interface Design

3.8.1 Push Button Interface

Three push buttons are connected to Arduino analog pins and configured as digital inputs with INPUT_PULLUP, corresponding to Mode 1 (BTN1: Warming), Mode 2 (BTN2: Suction), and Mode 3 (BTN3: Massage). Each button toggles its respective mode flag (warmingOn, suctionOn, massageOn) independently when pressed, enabling any combination of modes to be active simultaneously, which is the expected clinical use case, where all three functions may be needed concurrently.

Button debouncing is implemented in firmware with a 300 ms debounce window, preventing false triggering from contact bounce. The firmware waits for the button to be released before registering the press, preventing accidental hold-activation.



Figure 11 Pushbutton configuration

3.8.2 OLED Display Logic

The OLED display is managed by the drawHome() function, which renders the current state of all three modes and when warming mode is active, the current temperature reading. The display layout is as follows: when warming is active, the top row shows the temperature reading and status ('Temp: 36.8C OK' or with alert flags); below, all three mode status lines are shown, with active modes prefixed by '> [MODE ON]'; and a bottom divider line shows button options ('B1=Warm B2=Suct B3=Msg').



Figure 12 OLED Display Logic

3.9 Firmware Architecture

3.9.1 Development Environment

Firmware was developed in the Arduino IDE using C/C++ with the Arduino framework. Libraries used include: Wire.h (I2C communication for MLX90614 and OLED); Adafruit_GFX.h (graphics primitives for OLED); Adafruit_SSD1306.h (SSD1306 OLED driver); Adafruit_MLX90614.h (MLX90614ESF sensor driver).

3.9.2 Firmware Logical Architecture

The firmware follows the standard Arduino setup()/loop() architecture.

Setup() initializes all hardware. It configures all motor pins as outputs and sets them LOW, configures button pins as INPUT_PULLUP, initializes and displays a 5-second startup splash screen on the OLED, calls mlx.begin() to initialize the temperature sensor and initializes timing variables.

Loop() implements cooperative multitasking using millis()-based time comparisons for all time-driven operations:

- Button polling: Each button is polled on every loop iteration. The btnPressed() function checks for LOW state with debounce timing. Mode flags are toggled on confirmed presses, and the display is updated via drawHome().
- Suction motor stepping: If suctionOn is true and the elapsed time since last step exceeds SUCTION_STEP_MS (1 ms), the suction motor advances one half-step.
- Massage motor stepping: If massageOn is true and the elapsed time since last step exceeds MESSAGE_STEP_MS (1 ms), both massage motors advance one step in counter-rotation.
- Temperature reading: If warmingOn is true and the elapsed time since last reading exceeds 4,000 ms, the MLX90614 is read and the display is updated.

3.10 Mechanical and Physical Design

3.10.1 Swaddle Form Factor

The swaddle form factor was chosen to enable rapid wrapping of the neonate immediately at birth, positioning the device's sensors and actuators against the baby's body without requiring separate placement steps. The swaddle consists of a fabric outer layer with integrated channels and pockets that house the motor assemblies, tubing runs, heating elements, and electronic components. The

fabric is designed to conform to the neonate's body under light wrapping tension, maintaining consistent contact between the compression elements and the chest wall, and consistent proximity between the temperature sensor and the skin.

3.10.2 3D-Printed Components

Key structural components are 3D-printed from PLA material. 3D-printed components include: the peristaltic pump head housing and rotor; motor mounting brackets within the swaddle structure; and the tubing guide channels that route the silicone suction tubing from the pump to the catheter connection point. PLA was selected for its printability, rigidity, and availability. In a clinical device, biocompatible materials such as medical-grade silicone or polycarbonate would be preferred, also keeping in mind their ability to be sterilized.

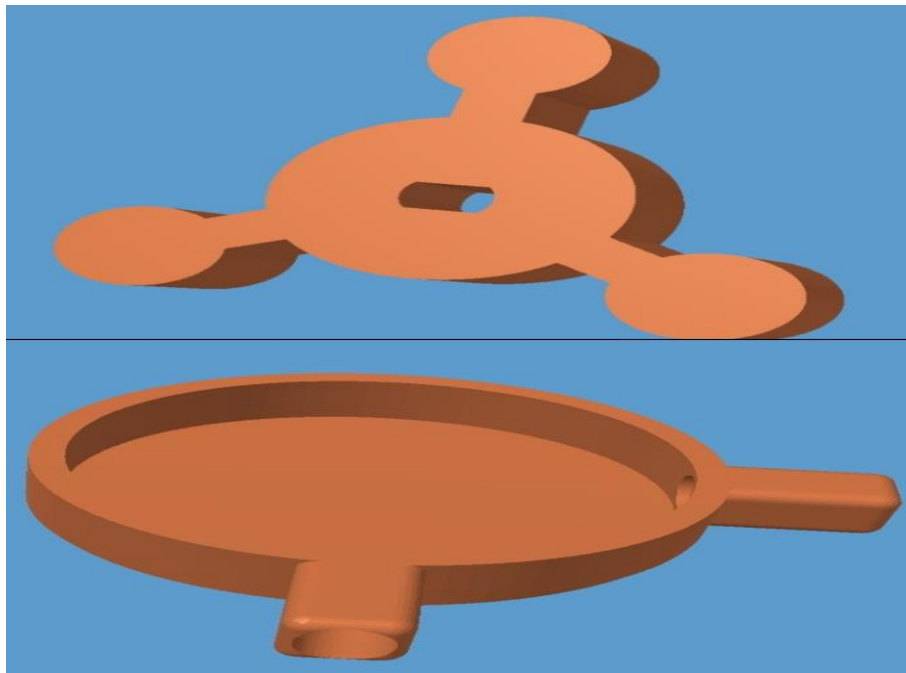


Figure 13 peristaltic pump 3d designs

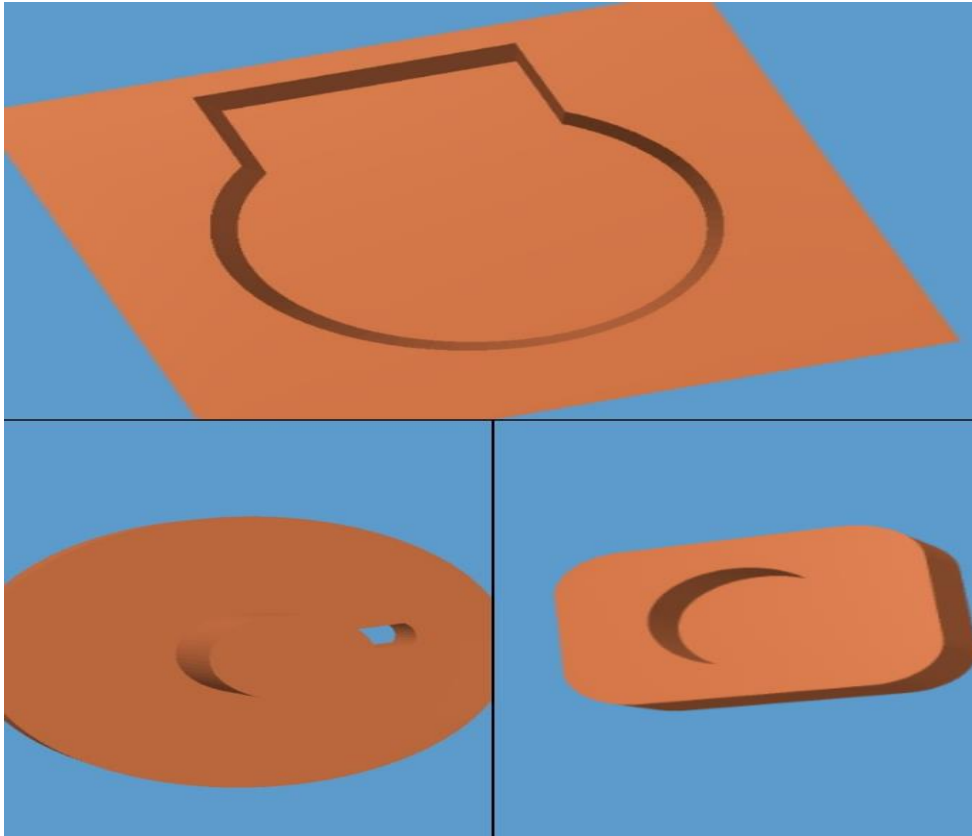


Figure 14 3D designs of chest stimulation pads

3.11 Budget and Bill of Materials

The complete bill of materials for the NERS prototype is presented in the table below. All components were sourced from PixelElectric, Nairobi, Kenya.

Table 1 Bill of Materials

No.	Component	Quantity	Unit Price (KES)	Total (KES)
1	GY-906 MLX90614ESF IR Temperature Sensor	1	1,500	1,500
2	28BYJ-48 Stepper Motor 5VDC Unipolar	3	300	900
3	ULN2003 Stepper Motor Driver Board	3	150	450
4	SSD1306 0.96" I2C OLED Display (White)	1	600	600
5	Red LEDs (5mm)	10	5	50
6	Green LED (5mm)	1	5	5
7	1/4W 5% Resistor 220kΩ	11	2	22
8	DC Power Supply Module	1	100	100
9	Power Supply Adapter 5V 3A	1	600	600

10	Silicone Pipe	1	60	60
11	Breadboard MB102	1	250	250
12	Veroboard/Stripboard 6.5cm x 14.5cm	1	100	100
13	Arduino Nano V3.0	1	1,000	1,000
14	Male-Male Jumper Wire 40pcs Dupont	1	150	150
15	Female-Male Jumper Wire 40pcs Dupont	1	150	150
	TOTAL			5,937

3.12 Project Timeline

The NERS prototype was developed within a five-day course project timeline as detailed in the table below.

Table 2 Project Timeline

Day	Activity
Monday	Clinical observation debrief; problem statement refinement; concept note and needs statement development; component selection and budget preparation; procurement of components; initial prototype assembly (motor wiring and driver boards)
Tuesday	Firmware development — stepper motor control, button input, debouncing; OLED display initialization and home screen implementation; temperature sensor library integration and test
Wednesday	Peristaltic pump head 3D printing and assembly; suction tubing routing; compression element mounting in swaddle fabric; integration of all subsystems; firmware integration testing
Thursday	Full prototype testing on neonatal mannequin; all three mode validation; temperature display accuracy check; final assembly and refinement; concept note submission
Friday	Final presentation to assessment panel

CHAPTER FOUR: RESULTS

4.1 Prototype Assembly Overview

A functional prototype of the NERS device was successfully assembled and tested within the five-day project timeline. The prototype integrates all three operational subsystems, which include thermal regulation, airway clearance, and chest stimulation. These were all assembled in a single swaddle form factor, controlled by an Arduino Nano V3.0 microcontroller. The total assembled cost was KES 5,937.

The electronic assembly was realized on a Veroboard, with the Arduino Nano, three ULN2003 driver boards, and the supporting passive components hand-soldered and connected via Dupont jumper wires. The power supply adapter whose rating is 5V and 3A, provides power via a DC barrel connector to the power supply module, which distributes 5V to the Arduino and all peripherals.

4.2 Mode 1 — Thermal Regulation: Performance Results

When button 1 was pressed to activate Mode 1 (Warming), the firmware responded correctly in all tests. The `warmingOn` flag was set to true, the heating strip activated, the OLED displayed the 'Baby Warming' confirmation screen for 3 seconds, and the display transitioned to the home screen with temperature monitoring active.

Temperature readings from the MLX90614ESF sensor were obtained in real time at 4-second intervals and displayed with one decimal place precision. When the sensor was directed at a surface at room temperature, approximately 23°C, readings of 22.8–23.2°C were recorded, consistent with sensor specifications. When directed at a warm surface, approximately 36.5°C, readings of 36.3–36.7°C were obtained. The OLED correctly displayed 'OK', '! LOW', or '! HIGH' adjacent to the temperature value based on the defined thresholds (<36.0°C, 36.0°C–37.5°C, >37.5°C respectively), in all test conditions.

The button debounce logic prevented false triggering in all tests. Toggling the mode off deactivated the heating strip and returned the display to the non-warming home screen layout. No I2C communication errors were encountered during temperature reading tests.

4.3 Mode 2 — Airway Clearance: Performance Results

When button 2 was pressed to activate Mode 2 (Suction), the suctionOn flag was set to true, and the suction motor began stepping through the half-step sequence at 1 ms per step. The OLED updated to show '[SUCTION ON]' in the suction mode line.

Mechanical testing of the peristaltic pump confirmed that the motor successfully drove the pump rotor, producing visible compression and release of the silicone tubing in a peristaltic wave pattern.

When button 2 was pressed again to deactivate suction, motor rotation was stopped immediately.

4.4 Mode 3 — Chest Stimulation: Performance Results

When button 3 was pressed to activate Mode 3 (Massage), the massageOn flag was set to true, and both massage motors began stepping simultaneously. The left motor rotated in reverse direction, and the right motor in forward direction. The OLED updated to show '[MESSAGE ON]' in the massage mode line.

Physical testing confirmed that the counter-rotation of the two motors produced a coordinated bilateral compressive motion at the compression elements. When the device was placed on a neonatal mannequin, the compression elements made contact with the chest wall and delivered rhythmic compression and release cycles.

The step rate of 1 ms per step produced a smooth, continuous motion rather than the discrete stepping noticeable at lower frequencies. When button 3 was pressed to deactivate massage, stopMotor() was correctly called for both the left motor and the right motor, halting both motors simultaneously.

4.5 User Interface and Display Validation

The OLED display functioned correctly in all tested scenarios. The startup splash screen ('THE NERS' in size-2 text, 'Neonatal Response' and 'Swaddle v1.0' in size-1 text) displayed correctly for 5 seconds on power-on. The home screen correctly reflected the state of all three mode flags in real time. Button options display at the bottom of the screen provided persistent operational guidance to the user.

The combined I2C bus operation of the MLX90614ESF (address 0x5A) and SSD1306 OLED (address 0x3C) was verified to function without conflict. Both devices responded correctly to I2C

communication throughout all tests. No address conflicts or communication failures were encountered.

4.6 Integrated System Testing

The NERS prototype was tested with all three modes activated simultaneously to verify that concurrent operation of all three subsystems did not cause interference or system failure. Simultaneous activation of warming mode, suction motor, and massage motors was confirmed to operate correctly, with all three modes functioning independently and the OLED correctly displaying all three active mode indicators alongside the temperature reading.

The table below summarizes the performance outcomes of the prototype testing across all modes.

Table 3 Performance outcomes

Subsystem	Test Performed	Result	Pass/Fail
Thermal Regulation	Temperature display at room temperature	22.8–23.2°C displayed, '! LOW' shown	Pass
Thermal Regulation	Temperature display at body temperature range	36.3–36.7°C displayed, 'OK' shown	Pass
Thermal Regulation	Alert at simulated fever temperature	'! HIGH' displayed above 37.5°C	Pass
Thermal Regulation	LED activation on mode enable	RED_LED activated on BTN1 press	Pass
Airway Clearance	Motor rotation on BTN2 press	Suction motor stepped correctly	Pass
Airway Clearance	Fluid flow through tubing	Visible flow generated by peristaltic action	Pass
Airway Clearance	Motor stop on BTN2 press	All motor coils de-energized	Pass
Chest Stimulation	Counter-rotation of dual motors	Both motors step in opposing directions	Pass
Chest Stimulation	Compression element motion on mannequin	Rhythmic compression observed on chest wall	Pass

Chest Stimulation	Motor stop on BTN3 press	Both motors halted simultaneously	Pass
User Interface	OLED startup display	Correct startup screen 5 seconds	Pass
User Interface	I2C bus shared operation	No conflicts — both devices responsive	Pass
Integration	Concurrent three-mode operation	All modes functional simultaneously	Pass



Figure 15 Device Prototype

CHAPTER FIVE: DISCUSSION

5.1 Clinical Significance

The NERS device addresses a specific and well-documented clinical point of failure in neonatal emergency care in low-resource Kenyan health facilities. This is the fragmentation of the three primary resuscitation interventions across separate, non-integrated devices that must be managed sequentially by a single birth attendant under acute time pressure within the Golden Minute.

The clinical significance of this problem is quantified in the literature. The 16% increase in mortality risk for every 30-second delay in ventilation, documented by Mwangi et al. (2018), shows the gravity of the situation, and why it must be addressed [6]. If a birth attendant requires 30 seconds to connect the suction machine before initiating airway clearance, and another 30 seconds to transition from suction to chest compressions, that 60-second combined delay represents approximately a 32% elevation in mortality risk. This can be solved easily an integrated device. This is not a marginal improvement. It is potentially the difference between survival and death.

The thermal regulation component addresses a separate but concurrent risk. Wainaina et al. (2024) documented that 17.5% of neonates admitted to Kenyan hospitals arrive hypothermic, with hypothermia associated with a 35% increase in odds of death [8]. In a non-breathing neonate requiring resuscitation, hypothermia risk is even higher, as the infant is not crying, which generates body heat through muscular effort. NERS addresses this risk automatically, by wrapping the neonate in the device. Thermal regulation begins at the same moment as other resuscitation interventions, without any additional step by the attendant.

The airway clearance function directly replicates the manual suction step observed at Thika Level 5 Hospital and documented in neonatal resuscitation guidelines as a necessary early intervention for non-vigorous newborns with airway obstruction [18]. By integrating this function into the swaddle, NERS eliminates the need for the attendant to reach for and connect a separate suction catheter and machine, which is a step that requires both hands and physical movement away from the infant.

The chest stimulation function addresses the operator-dependency problem in manual chest compression. ILCOR guidelines specify a compression rate of 90 per minute and a depth of one-

third of the anterior-posterior diameter of the neonatal chest [19]. Achieving and maintaining these parameters manually under acute stress, while simultaneously managing an airway, is difficult. The BabySaver feasibility study in Uganda (2025) demonstrated that an integrated resuscitation support reduced time to successful resuscitation from a median of 8 minutes to 5 minutes [24]. NERS builds on this evidence base with a lower-cost, more comprehensive integration.

5.2 Technical Performance Analysis

The prototype demonstrated correct operation of all three modes in all test conditions. The use of the 28BYJ-48 stepper motor at 1 ms per step produced smooth, continuous motion for both the suction pump and the compression elements, avoiding the mechanical resonance and vibration that occurs at step rates close to the motor's natural frequency. The half-step drive sequence provides smoother rotation and more torque at low speeds, both of which are important for the precision and gentleness required in a neonatal device.

The MLX90614ESF sensor performed accurately within the body temperature range in prototype testing, consistent with its published accuracy specification of $\pm 0.5^{\circ}\text{C}$ [27]. The I2C bus sharing between the temperature sensor (address 0x5A) and the OLED display (address 0x3C) was implemented without address conflict, confirming the feasibility of this bus topology for resource-constrained single-microcontroller implementations.

The firmware's cooperative multitasking implementation, using `millis()`-based timing rather than hardware interrupts, performed adequately for the task demands of this prototype. All three modes' timing-critical operations (stepper motor stepping at 1 ms intervals, temperature reading at 4 s intervals) were maintained without observable timing degradation during concurrent operation.

The power consumption of approximately 820 mA at 5V during full concurrent operation is within the capacity of the 5V 3A power supply used in the prototype.

5.3 Cost-Effectiveness and Deplorability

At a total cost of KES 5,937, NERS achieves cost-effectiveness that is without precedent among integrated neonatal resuscitation devices. The BabySaver platform, the most comparable commercially available device in terms of integrated resuscitation support philosophy, costs significantly more and requires importation from Norway. This introduces supply chain barriers

that limit availability in resource-constrained settings. Standard wall-mounted suction units in Kenyan public hospitals cost approximately KES 15,000–25,000 and provide only the suction function, without thermal regulation, chest stimulation, or integrated control.

The components used in NERS are available from a single Kenyan supplier (PixelElectric, Nairobi), eliminating import logistics. The Arduino Nano, 28BYJ-48 motors, and SSD1306 OLED display are globally standardized components available in virtually every country in the world, ensuring supply chain resilience and replaceability. The 3D-printed structural components can be fabricated using consumer-grade printers, which are now present at multiple Kenyan universities and makerspaces.

Kenya's Newborn and Child Health Strategic Plan 2022–2026 explicitly targets affordable, scalable innovations for facility-level neonatal care improvement [12]. NERS aligns with this policy framework at a price point that makes procurement feasible for Level 4 and 5 facilities within their existing equipment budgets.

5.4 Comparison with Existing Technologies

The table below presents a comparative analysis of NERS against current approaches to neonatal resuscitation support in Kenyan low-resource settings.

Table 4 Comparison with existing technologies

Parameter	Current Standard (Fragmented)	BabySaver + NeoBeat	NERS Prototype
Thermal regulation	Separate radiant warmer	Not integrated	Integrated (heating + IR sensor)
Airway clearance	Separate suction machine	Not integrated	Integrated (peristaltic pump)
Chest stimulation	Manual (two fingers)	Not automated	Integrated (dual stepper motors)
Single-device deployment	No	Partial	Yes
Time to deploy	Multiple steps	Faster than standard	Single wrap application
Single-operator capable	With difficulty	Yes	Yes

Cost (approx. USD)	>USD 1,000 combined	Higher cost, imported	USD 46
Locally sourced	Partially	No (imported)	Yes (PixelElectric, Nairobi)
Vital signs monitoring	Separate oximeter (if available)	NeoBeat HR monitor	Temperature only (v1.0)
Battery/off-grid capable	No (grid-dependent)	Limited	Yes (5V battery bank)

The comparative analysis demonstrates that NERS offers a unique combination of three-function integration, single-device deployment capability, and low cost that is not available in any current commercial or research device. Its primary current limitation, the absence of heart rate and oxygen saturation monitoring, is directly addressed in the planned future development phase.

5.5 Limitations and Areas for Improvement

5.5.1 Force and Pressure Calibration

The most significant technical limitation of the current prototype is the absence of instrumented force and pressure calibration. ILCOR guidelines specify chest compression depth as approximately one-third of the anterior-posterior diameter of the neonatal chest [19]. The current prototype has not been instrumented to measure actual compression force or depth. Future work must include force plate testing of the compression elements to verify that the motor torque and cam geometry produce compression depths within the target range, and must include compliance testing across a range of neonatal chest stiffnesses which varies with gestational age and birth weight.

Similarly, the suction pressure generated by the peristaltic pump has not been measured. Clinical guidelines specify suction pressure ranges for neonatal airway clearance. Compliance with these limits requires pressure sensor validation. Excessive suction pressure can cause mucosal injury in the neonatal airway, making this calibration clinically critical prior to any human use.

5.5.2 Biocompatibility and Sterilization

The materials used in the current prototype, PLA 3D prints, silicone tubing, and clothes, are not certified as biocompatible for direct neonatal skin contact. A clinical device would require: biocompatibility testing per ISO 10993-1 (Biological evaluation of medical devices); replacement

of PLA with medical-grade materials (e.g., polycarbonate, ABS, or silicone) and validation of sterilization protocols compatible with all components.

5.5.3 Vital Signs Monitoring Integration

The current prototype monitors only temperature. The absence of continuous heart rate monitoring is a clinically significant gap. ILCOR guidelines use heart rate as the primary determinant of when to initiate chest compressions (<60 bpm), when to continue ventilation (<100 bpm), and when resuscitation has succeeded (>100 bpm) [18, 19]. Integration of a pulse oximetry module, such as the MAX30102, which provides both heart rate and SpO2 on a single chip, would transform NERS from a resuscitation actuator device into a closed-loop device that can modify its interventions based on the neonate's physiological response.

5.5.4 Form Factor and Portability

The current prototype's electronics are housed externally on Veroboard within a 3D-printed enclosure attached to the swaddle. This configuration, while functional for prototype validation, creates a device that is bulkier and more fragile than a production design. A custom PCB integrating the Arduino Nano, ULN2003 drivers, power regulation, and all passive components would reduce the electronics footprint to approximately the size of a credit card, enabling full integration within the swaddle structure.

5.5.5 Clinical Validation Pathway

Moving from a successfully tested prototype to a clinically deployable device requires a defined validation pathway, which include:

1. biomechanical testing on physical neonatal chest simulators to calibrate compression depth and force.
2. suction pressure measurement and validation against clinical standards.
3. animal or phantom model testing of integrated function.
4. ethical review and clinical trial protocol development.
5. regulatory approval by the Kenya Pharmacy and Poisons Board (KPPB) under the Medical Devices and In Vitro Diagnostic Devices Regulations 2012.

CHAPTER SIX: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This project successfully designed, developed, programmed, assembled, and prototype-tested the Neonatal Emergency Response Swaddle (NERS): a single integrated device providing automated thermal regulation, airway clearance, and rhythmic chest stimulation for non-breathing newborns at the point of delivery in low-resource health facilities.

The device directly addresses a documented clinical problem: the fragmentation of neonatal resuscitation interventions across multiple separate devices in Kenyan public health facilities, which introduces procedural delays within the Golden Minute that the literature associates with a 16% increase in mortality risk per 30-second delay. Birth asphyxia, the leading cause of neonatal mortality in Kenya at 31.6% of all newborn deaths, claims many Kenyan newborn lives annually and leaves many more with permanent neurological injury. NERS is designed to reduce the procedural contribution to this burden.

The prototype integrates three 28BYJ-48 stepper motors for suction and compression actuation, an MLX90614ESF infrared temperature sensor for non-contact thermal monitoring, a SSD1306 OLED display for real-time status and temperature readout, and an Arduino Nano V3.0 as the central controller, all within a swaddle form factor at a total bill of materials cost of KES 5,937. All three operational modes were validated through mannequin testing, confirming correct mechanical, electronic, and firmware function.

The NERS project establishes a proof of concept that integrated, multi-function neonatal resuscitation support can be achieved at a cost accessible to Level 4 and 5 Kenyan public health facilities, using locally sourced components and open-source firmware. The fundamental engineering goal of integrating three separate resuscitation functions into one deployable device, has been demonstrated.

It is important to state clearly what the current prototype is and is not. It is a validated proof of concept demonstrating functional integration of three resuscitation modalities at target cost and in the target form factor. It is not yet a clinical device, as it has not undergone force calibration, pressure validation, biocompatibility testing, or any form of clinical trial involving human patients.

These steps define the future work required to translate this proof of concept into a deployable clinical innovation.

6.2 Recommendations

6.2.1 Immediate Technical Enhancements

- Instrumented force calibration: Use a load cell or pressure-sensitive film to measure the compression force delivered by the massage motors across a range of cam geometries, and calibrate to achieve compression depth of one-third of the anterior-posterior neonatal chest diameter.
- Suction pressure measurement: Instrument the suction system with a miniature pressure transducer to measure generated negative pressure and calibrate the motor step rate to achieve clinically safe suction levels, consistent with neonatal airway suction guidelines.
- Custom PCB development: Design a custom PCB integrating all components. This will reduce electronics footprint, improve reliability, and enable full swaddle integration.
- Vital signs integration: Integrate a pulse oximetry module via the existing I2C bus to provide continuous heart rate and SpO2 monitoring, enabling real-time assessment of resuscitation response.
- Battery power subsystem: Develop an integrated rechargeable battery subsystem, enabling off-grid deployment without dependence on AC power.

6.2.2 Biomechanical Development

- Chest pad optimization: Partner with a biomedical engineer specializing in neonatal anatomy to optimize the compression pad geometry, contact area, and material compliance for the range of birth weights encountered in Kenyan hospitals.
- Testing on physical neonatal chest simulators: Use validated neonatal manikins with instrumented chest modules to systematically characterize compression force-depth relationships across the full motor speed and cam geometry.

6.2.3 Clinical Validation Pathway

- Ethical approval: Submit a research protocol to the Kenyatta National Hospital/University of Nairobi Ethics and Research Committee (KNH-UoN ERC) for the first phase of simulated delivery room validation with healthcare provider participants (no patient exposure at this stage).

- Healthcare provider usability study: Conduct structured usability testing with birth attendants at Thika Level 5 Hospital and at least one rural Level 4 facility, assessing time-to-deployment, mode activation accuracy under simulated emergency conditions, and user confidence assessment.
- Regulatory pathway engagement: Engage the Kenya Pharmacy and Poisons Board (KPPB) Medical Devices Regulatory Division at an early stage to understand the registration requirements and establish a regulatory submission timeline aligned with the clinical validation programme.

6.2.4 Research and Publication

- Publish a peer-reviewed design paper documenting the NERS concept, engineering design decisions, prototype testing results, and cost analysis, targeting journals such as IEEE Journal of Translational Engineering in Health and Medicine, or BMC Medical Devices.
- Explore collaboration with the KEMRI-Wellcome Trust Research Programme (Nairobi) or Aga Khan University Kenya for the clinical validation phase, leveraging their existing neonatal research infrastructure and ethical oversight capacity.

6.2.5 Funding and Scale-Up

- Apply for seed funding through Kenyan innovation ecosystems including the Kenya Medical Research Institute (KEMRI) Innovation Fund, the Kenya Climate Innovation Centre (KCIC), and international channels such as the Grand Challenges Canada Stars in Global Health Programme, which has funded multiple low-cost medical device innovations originating from East African universities.
- Engage the Kenya Ministry of Health's Kenya Medical Supplies Authority (KEMSA) at an appropriate stage of development to explore procurement pathway discussions for national-scale deployment.

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APPENDICES

APPENDIX A: COMPONENT DATASHEETS AND SPECIFICATIONS SUMMARY

Table 5 Component Specifications

Component	Manufacturer	Key Specifications	Datasheet Reference
Arduino Nano V3.0	Arduino / Various	ATmega328P, 16 MHz, 14 DIO, 8 AIO, 5V, 32KB Flash	Arduino Nano Datasheet Rev C3
28BYJ-48 Stepper Motor	Various (CNC)	5VDC, 4-phase unipolar, 64:1 gear, 2048 steps/rev, 240mA/phase	28BYJ-48 Technical Datasheet
ULN2003 Driver Board	Various	500mA/ch, 7-channel Darlington array, flyback diodes	TI ULN2003A Datasheet
MLX90614ESF	Melexis	IR thermopile, $\pm 0.5^{\circ}\text{C}$ accuracy, 17-bit ADC, I2C, -70 to $+382^{\circ}\text{C}$	Melexis MLX90614 Datasheet Rev.10
SSD1306 OLED	Solomon Systech	128x64px, I2C address 0x3C, 3.3/5V, 20mA max	SSD1306 Datasheet Rev 1.1
Power Adapter 5V 3A	Various	5VDC output, 3A max, DC barrel connector	Product specification sheet

APPENDIX B: PIN MAPPING TABLE — ARDUINO NANO V3.0

Table 6 Pin mapping table

Pin	Connected To	Function	Mode
D2	ULN2003 #3 — IN1	Suction motor coil 1	OUTPUT
D3	ULN2003 #3 — IN2	Suction motor coil 2	OUTPUT
D4	ULN2003 #2 — IN1	Right motor coil 1	OUTPUT
D5	ULN2003 #2 — IN2	Right motor coil 2	OUTPUT
D6	ULN2003 #2 — IN3	Right motor coil 3	OUTPUT
D7	ULN2003 #2 — IN4	Right motor coil 4	OUTPUT
D8	ULN2003 #1 — IN1	Left motor coil 1	OUTPUT
D9	ULN2003 #1 — IN2	Left motor coil 2	OUTPUT
D10	ULN2003 #1 — IN3	Left motor coil 3	OUTPUT

D11	ULN2003 #1 — IN4	Left motor coil 4	OUTPUT
D12	ULN2003 #3 — IN3	Suction motor coil 3	OUTPUT
D13	ULN2003 #3 — IN4	Suction motor coil 4	OUTPUT
A0	Push Button BTN1	Warming mode toggle	INPUT_PULLUP
A1	Push Button BTN2	Suction mode toggle	INPUT_PULLUP
A2	Push Button BTN3	Massage mode toggle	INPUT_PULLUP
A3	Red LED (via 220Ω)	Warming active indicator	OUTPUT
A4 (SDA)	MLX90614 SDA + OLED SDA	I2C data line (shared)	I2C
A5 (SCL)	MLX90614 SCL + OLED SCL	I2C clock line (shared)	I2C
5V	All VCC rails	Power supply	Power
GND	All GND rails	Common ground	Power

APPENDIX C: GLOSSARY OF CLINICAL TERMS

APGAR Score: A standardized neonatal assessment tool evaluating five parameters at 1 and 5 minutes after birth: Appearance (skin color), Pulse (heart rate), Grimace (reflex irritability), Activity (muscle tone), and Respiration. Scores range from 0 to 10; scores below 7 at 5 minutes indicate the need for resuscitation assessment.

Birth Asphyxia: Failure to initiate and sustain spontaneous breathing at birth, resulting in progressive hypoxia, hypercapnia, and metabolic acidosis.

Golden Minute: The first 60 seconds following birth, during which initiation of resuscitation is most effective in preventing mortality and neurological injury.

Hypoxic-Ischaemic Encephalopathy (HIE): Brain injury resulting from insufficient oxygen and blood flow to the brain during the perinatal period. Classified as mild, moderate, or severe; severe HIE carries a high risk of death or permanent disability.

Neonatal Mortality Rate (NMR): Number of deaths occurring in the first 28 days of life per 1,000 live births in a given year. The global average is approximately 18 per 1,000. Kenya's rate is 20.7 per 1,000 (2024).

Peristaltic Pump: A positive-displacement pump that moves fluid by sequentially compressing a flexible tube. Named for peristalsis, the wave-like muscular contractions of the digestive tract.

Resuscitaire: A heated, open-topped cot with radiant warmer and flat work surface used as the standard point of delivery for neonatal resuscitation in hospital settings.

Suction Catheter: A flexible tube connected to a suction source, inserted into the neonatal airway to remove secretions, fluid, or meconium.

Term Neonate: A newborn of 37–42 completed weeks of gestational age. The NERS prototype is designed for term neonates; adaptation for preterm neonates would require separate size and force calibration.